A Self-starting Boost Converter with Multi-Mode Operation and Direct-Sensing MPPT for Photovoltaic Energy Harvesting

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***Abstract*—This paper presents a self-starting boost converter designed for photovoltaic energy harvesting. Multi-mode operation accelerates circuit startup, allowing the interface to efficiently reach its operating voltage within a very short time. A direct-sensing maximum power point tracking (MPPT) technique is introduced to improve power conversion efficiency across a wide illumination range. In addition, a periodic sleep mode is incorporated to reduce overall power consumption while maintaining high MPPT tracking accuracy. Fabricated in a 180 nm CMOS process, the proposed circuit achieves self-startup within 72.8 ms at an input voltage of 180 mV, with a peak tracking efficiency of 99.6% and an end-to-end efficiency of 90.6%. Compared to state-of-the-art designs, the proposed solution offers a better balance between power consumption and tracking performance.**

***Index Terms*—Energy harvesting, maximum power point tracking, self-startup, boost converter, power consumption**

I. Introduction

Conventional Internet of Things (IoT) nodes predominantly rely on battery power, this solution exhibits three inherent drawbacks: limited operational lifespan, potential environmental contamination risks, and substantial maintenance challenges [1]. To address these challenges, energy harvesting technologies have emerged as a promising solution by capturing and converting ambient energy into usable electrical power. Among various renewable energy sources, light energy stands out as particularly advantageous for powering IoT nodes, owing to its superior energy density and reliability compared to other ambient energy sources [2]. However, the output power of photovoltaic (PV) cells exhibits nonlinear power-voltage relationships that are dependent on environmental conditions (particularly solar irradiance), semiconductor doping profiles, and atmospheric factors [3]. To maximize energy harvesting from renewable sources, an effective maximum power point tracking (MPPT) method is essential.

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Recent studies have demonstrated various MPPT methods, including discontinuous and continuous techniques. The Fractional Open-Circuit Voltage (FOCV) method is widely used due to its simplicity, low power overhead, and compatibility with different converter controls [4]. However, it operates discontinuously by requiring periodic disconnection of the source for voltage sampling, and the ratio of maximum power point voltage (V_{MPP}) to open-circuit voltage (V_{OC}) varies with irradiance, typically ranging between 0.65 and 0.8 for most PV cells [5]. In contrast, continuous MPPT techniques such as Perturb and Observe (P&O) maintain real-time regulation without power interruption, enabling fast and accurate tracking under dynamic conditions [6-8]. Yet conventional continuous methods often rely on power-hungry detection circuitry. For example, [6] used a complex current sensing circuit, resulting in 3.4 mW quiescent power, while [7] employed a linear analog multiplier to compute input power, reducing quiescent power to 25 μW. [8] implemented a continuous tracking method, but achieving only 96.2% tracking efficiency.

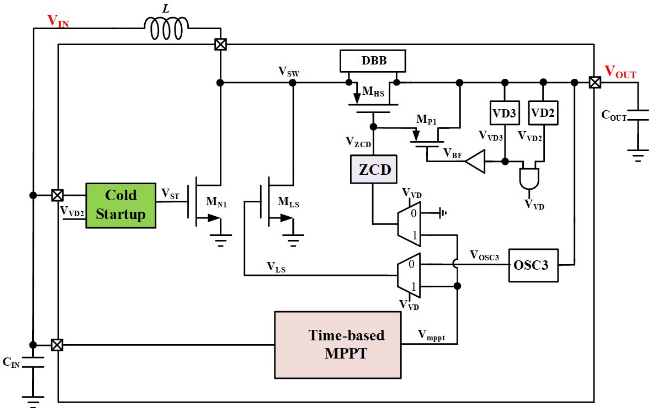


Fig. 1 System block diagram of the proposed boost converter.

To overcome these issues, this work proposes a time-based MPPT approach with continuous tracking capability. The system directly measures the PV cell's input power using a low-power time-domain multiplier and incorporates a periodic sleep mode to turns off main circuits after reaching MPP. This design achieves high tracking accuracy and converter efficiency. Furthermore, it integrates a cold-start block for self-startup operation.

II. The Proposed Boost Converter

Fig.1 shows the proposed boost converter system, comprising a cold-start block, a time-based MPPT block, and a power stage. For efficient low-voltage operation, it operates in three modes: cold-startup, auxiliary boost, and main boost. A time-based MPPT controller adjusts on-time for source impedance matching, and includes a sleep mode to minimize MPPT circuit power. A low-power, high-precision digital ZCD circuit prevents reverse current from V_{OUT} to V_{SW} by adjusting M_{P1}'s on-time to maximize output voltage.

A. Operating Modes of the Proposed Boost Converter

Fig. 2 (a) shows the block diagram for cold startup. First, the input voltage V_{IN} is boosted by a cross-coupled linear charge pump (CP), which is clocked by a stacked-inverter ring oscillator (OSC1). When the output of the CP, V_{CP}, rises to 600 mV, the output of voltage detector, V_{VD1}, goes high and enables the auxiliary oscillator (OSC2). The oscillator, powered by V_{CP}, generate a pulse signal V_{ST} to turn on the low-side switch, M_{N1} in auxiliary boost mode. Meanwhile, V_{in} is controlled by the voltage detector output V_{VD2}.

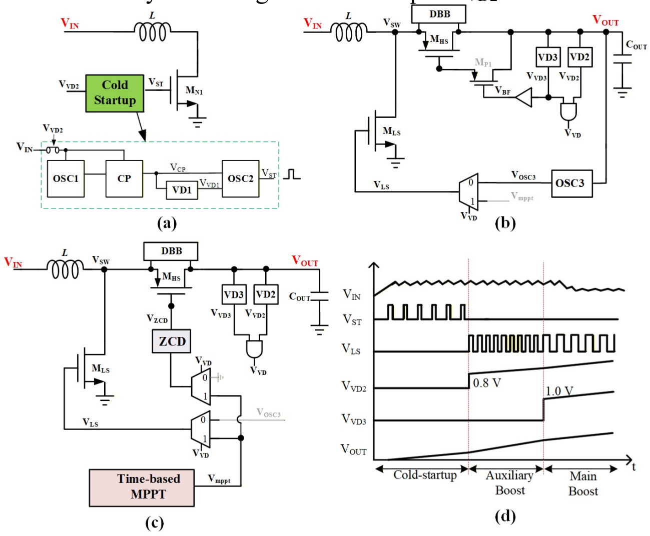


Fig. 2 Block diagram of the proposed boost converter in (a) cold-startup, (b) auxiliary boost, (c) main boost mode, and (d) its operating waveform

Fig. 2 (b) shows the auxiliary boost mode, which begins by charging L via M_{N1} triggered by V_{VD1}. At the falling edge of V_{VD1}, the inductive overshoot at V_{SW} turns on the diode-connected switch M_{HS}, transferring power to storage capacitor, C_{OUT} in asynchronous operation. Once V_{OUT} exceeds 800 mV, the output of voltage detector (VD2), V_{VD2}, becomes high, thereby selecting the output signal of the third oscillator (OSC3) V_{osc3} to provides a higher clock frequency for continue operation. Now, the energy transfer is performed with a wider low-side switch, M_{LS} and a higher switching frequency signal, V_{LS}, accelerating the overall energy accumulation process. Meanwhile, the V_{VD2} disables the cold startup block for avoiding unnecessary power consumption.

Fig. 2 (c) shows the main boost mode. When V_{OUT} exceeds 1.0 V, V_{VD3} outputs high, connecting V_{mppt} (from the MPPT block) to the gate terminal of M_{LS} and the ZCD controller. M_{P1} is turned off, while the high side switch M_{HS} is regulated by the ZCD controller, enabling efficient synchronous boost mode for high output power. The bulk voltage of M_{HS} is dynamically biased to the highest potential using a dynamic body biasing (DBB) technique for avoiding body current. The corresponding operating waveform is shown in Fig. 2 (d).

B. MPPT Method

To enhance the energy harvesting efficiency, a direct MPPT method is adopted. As shown in Fig. 3, the MPPT block consists of a power monitor and an impedance controller. The power monitor senses the average voltage V_{AVG} and average current I_{AVG} from the PV unit and multiplies them with a time multiplier, as shown in (1) to obtain the input power (P_{IN}) information. The output of power monitor adjusts the input impedance of the boost converter (Z_{IN}) for extracting PV power as much as possible.

$$
\begin{aligned}
P_{IN} &= I_{AVG} \times V_{AVG} = C_{IN} \frac{V_R}{T_{\text{con}}} \times V_{AVG} \\
&= C_{IN} \frac{\left(V_{PK} - V_{VLY}\right)}{T_{\text{con}}} \times V_{AVG}
\end{aligned}
\tag{1}
$$

where V_R represents the voltage ripple of V_{IN}, with V_{PK} and V_{VLY} denoting the peak and valley values of V_{IN}, respectively. T_{con} is the operating period of the MPPT circuit. The results demonstrate that the P_{IN} information can be derived by sensing and multiplying V_{PK}-V_{VLY} and V_{AVG}.

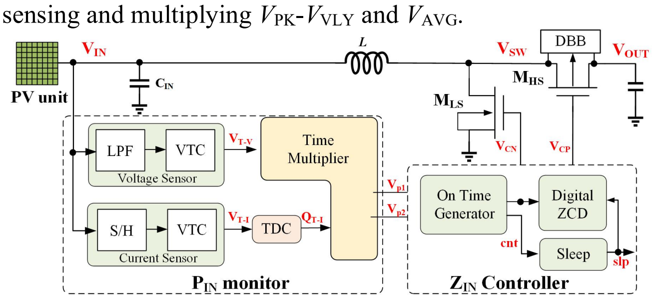


Fig. 3 Block diagram of the proposed MPPT circuit

In the proposed power monitor, a low-pass filter and a sample & hold (S/H) circuit have been employed, respectively, to extract V_{AVG} and I_{AVG} information from the PV unit. With a voltage-to-time converter (VTC), these two parameters are further converted into variables $V_{T\text{-}V}$ and $V_{T\text{-}I}$, respectively. The pulse width of $V_{T\text{-}V}$ and $V_{T\text{-}I}$ are Δt_V and Δt_I, respectively, which represent V_{AVG} and I_{AVG}. A time-to-digital converter (TDC) is then used to converter the time information in $V_{T\text{-}I}$ into a 48-bit thermometer code ($Q_{T\text{-}I}$). The time multiplier generates the output signal of V_{p1} and V_{p2}, which are the results of $V_{AVG} \times I_{AVG}$. at different moments.

The on-time generator in the impedance controller compares the input power at the current moment (V_{p1}) with that at the previous moment (V_{p2}) and determine the variation of the on time (T_{on}) of power FET M_{LS} and M_{HS}; if the present operating point is on the left side of the maximum power point (MPP) and $V_{p1} > V_{p2}$, as shown in Fig.4, T_{on} is reduced and the operating point moves to the right towards the MPP. Conversely, if the present operating point is on the right side of the MPP and $V_{p1} < V_{p2}$, the operating point moves to the left towards the MPP. The location of the present operating point is captured by comparing V_{AVG} and P_{IN} at the current moment with those at the previous moment.

A sleep mode is integrated into the impedance controller to lower the power consumption of the MPPT circuit. As shown in Fig.3, when the comparison result (cnt) between V_{p1} and V_{p2} exhibits the pattern of "1→0→1→0" or "1→1→0→0→1→1" the operating point is identified as being near the MPP. The MPPT circuit is then disabled and enters sleep mode for a predefined duration. To prevent the reverse current from V_{OUT} to V_{SW}, a digital zero-current detector (ZCD) is adopted. The ZCD consists of of a chain of digital logic units and operates with very low static power consumption.

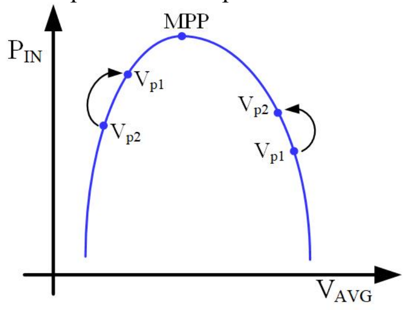


Fig. 4 Variation of MPP

C. Power monitor

Fig. 5 shows the schematic and timing diagram of the proposed voltage-to-time converter (VTC) in the proposed power monitor. The conversion can be divided into two phases. In Φ_V phase, comparator CMP1 is enabled and comparing the V_{AVG} with the ramp voltage V_{RAMP}, which is generated by charging/discharging C_{RAMP} with bias current I_B. Initially, V_{TV} is at a high level. When the V_{RAMP} exceeds V_{AVG}, V_{TV} switches to a low level. In Φ_I phase, comparator CMP2 and CMP3 are enabled and the V_{RAMP} is compared with V_{PK} and V_{VLY}. The pulse width of V_{TV} ($\triangle t_V$) and V_{TI} ($\triangle t_I$) can be represented as,

$$\Delta t_V = \frac{V_{AVG}}{V_{PK,RAMP}} \cdot T_{con} \tag{2}$$

$$\Delta t_I = \frac{V_{PK} - V_{VLY}}{V_{PK,RAMP}} \cdot T_{con} \tag{3}$$

where T_{chg} is the operating period of energy extraction and $V_{PK,RAMP}$ is the peak voltage of V_{RAMP}.

A delay-line based time-to-digital converter is employed to converted the pulse width of $\triangle t_I$ into 48-bit thermometer code,

$$Q\,[47:0] = \Delta t_I / t_{delay} \tag{4}$$

where t_{delay} is the transmission delay of single D flip-flop.

Fig.6 shows the schematic of the time multiplier, in which Q [47:0] is applied to the binary weighted current. Thus, the current I_{TI} can be given by,

$$I_{TI} = Q[47:0] \cdot I_{UNIT} \tag{5}$$

where I_{UNIT} is the unit current flowing through P_{48}.

Assuming that the circuit operates at Φ_1 phase, the current I_{TI} charges the capacitor $C_{\Phi 1}$ for time duration $\triangle t_V$ and the value of V_{P1} can be written as,

$$V_{P1} = \frac{I_{TI} \cdot \Delta t_V}{C_{\phi 1}} = \frac{\Delta t_I}{t_{delay}} \cdot \frac{I_{UNIT} \cdot \Delta t_V}{C_{\phi 1}} \tag{6}$$

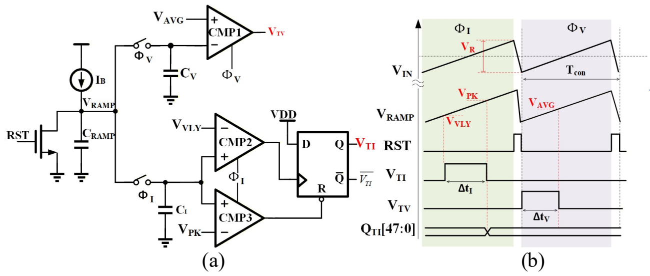


Fig. 5 (a) Schematic and (b) timing diagram of the proposed voltage-to-time converter

Substituting (2) and (3) into (6), V_{P1} can be further given by,

$$\begin{aligned} V_{P1} &= \frac{I_{UNIT}}{t_{delay} C_{\phi 1}} \cdot \Delta t_I \cdot \Delta t_V \\ &= \frac{I_{UNIT}}{t_{delay} C_{\phi 1}} \cdot V_{AVG} \cdot \left(V_{PK} - V_{VLY}\right) \cdot \left(\frac{T_{chg}}{V_{PK,RAMP}}\right)^2 \end{aligned} \tag{7}$$

The value of V_{P2} can be obtained with the same procedure. Based on the above results, the power information at current and previous phase can be represented and compared for maximum power extraction.

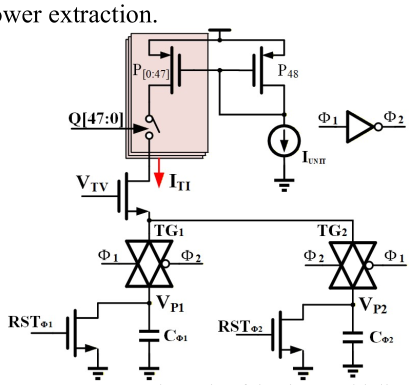


Fig. 6 Schematic of the time multiplier

D. Impedance Controller

Fig. 7 shows the on-time generator in the proposed impedance controller. The V_{AVG} comparator and V_P comparator compare the current and previous V_{AVG} and input power, respectively, generating UP/DN signals. If the current operating point locates at the left of the MPP, UP/DN goes low, reducing delay cell's delay and shortening t_{on}. The shorter t_{on} decrease I_{AVG} and increase V_{AVG}, making the operating point rightward toward the MPP. The signal UP/DN is transmitted to a 8 bit register. When the Sleep logic cell detects the UP/DN signal is 10101010 or 11110000, it sets the signal *SLEEP* high, indicating that the circuit is currently near the MPP. After a predefined sleep time set by CLK2, the circuit resumes normal operating.

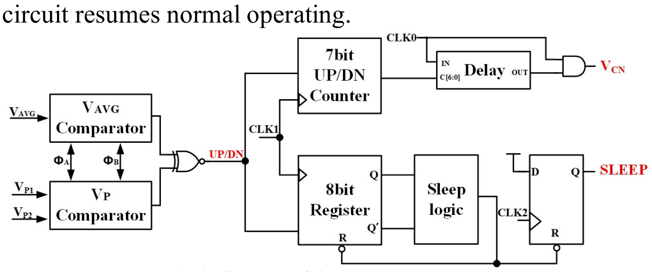


Fig. 7 Block diagram of the proposed on-time generator

A flip-flop-based sensing technique is employed to track the zero point of inductor current I_L, as shown in Fig. 8. Assuming the inductor energy is not fully transferred to the output load when the high-side PMOS transistor is turned off at the rising edge of signal V_{CP}, thus, the voltage V_{SW} shows an overshoot. The high overshoot voltage is transferred to the later 7-bit counter at the rising edge of $V_{CP,LS}$. The resulting output increases the delay between the falling edge of V_{CN} and $V_{CN,LS}$ via a programmable delay cell (Delay) and create a longer on-time in the next cycle for high-side PMOS transistor. The procedures are also the case when the inductor energy is fully transferred to the load and reverse current has formed. Fig. 8 illustrated the timing diagram of the digital ZCD.

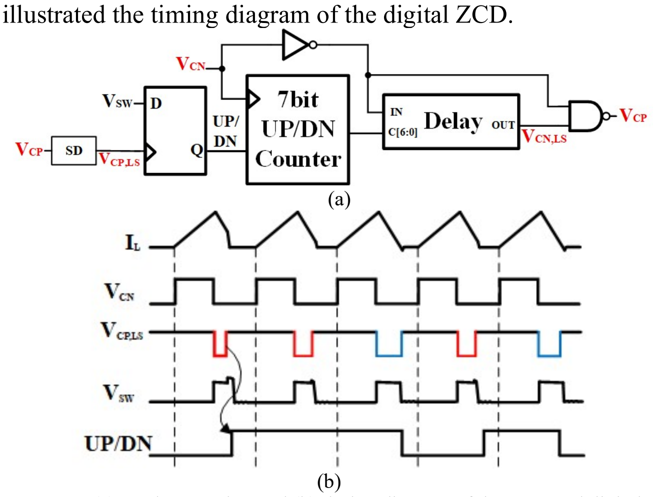


Fig. 8 (a) Implementation and (b) timing diagram of the proposed digital ZCD

III. MEASUREMENT RESULTS

The proposed interface circuit is fabricated using a 180 nm CMOS process with an active chip area of 1.67 mm^2. Fig. 9 shows the chip micrography and simulated current breakdown of the proposed circuit. The total current consumption is 1.134 µA. For the measurement setup, the prototype is connected to a photodiode (CRM ICRO PD60), which is illuminated by a cold light source (WC-XD 301). The 1 illumination is then measured using an illuminometer (Smart sensor AS823).

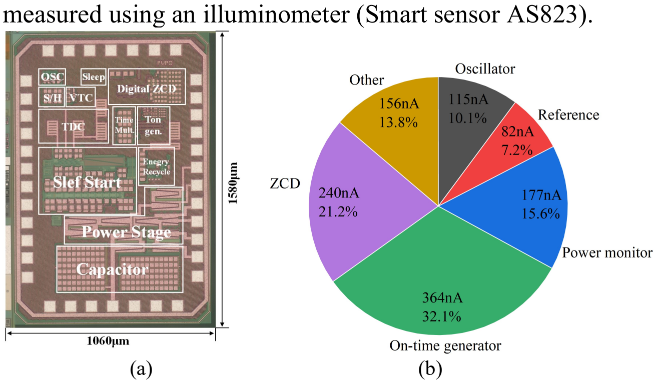


Fig. 9 (a) Chip Micrograph and (b) circuit consumption of the proposed energy harvester.

Fig. 10 shows the cold start of the interface circuit at V_{IN} = 0.18 V. As shown, V_{OUT} reaches 0.81 V after 63 ms, completing cold start. The circuit then enters asynchronous boost mode, with OSC3 outputting a higher frequency clock to drive low-side transistor M_{LS}. V_{OUT} reaches 0.99 V after 9.8 ms. At this point, the circuit switches to synchronous boost mode, with the MPPT module driving M_{LS} via V_{mppt}.

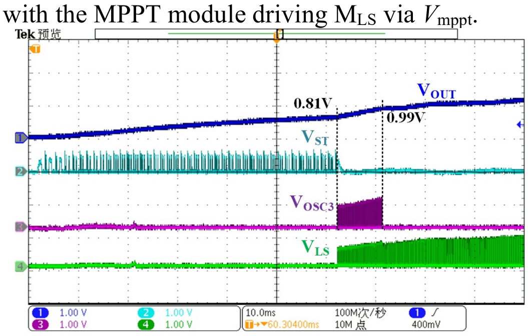


Fig. 10 Measured cold start process of the proposed interface circuit

Fig. 11 shows the MPPT process of the interface circuit under two illumination conditions: 5000 and 20,000 Lux. As illustrated in Fig. 11(a), at low illumination, V_{IN} is approximately 0.30 V. At this point, the output power at the previous moment V_{P1} and the output power at the current moment V_{P2} reach the same value, indicating that the circuit enters the MPPT state. After a sustained period, the sleep signal *SLEEP* goes high, and the entire circuit enters sleep mode. The circuit re-enters MPPT mode after 36 ms. The similar operation is observed under high illumination, as shown in Fig. 11 (b).

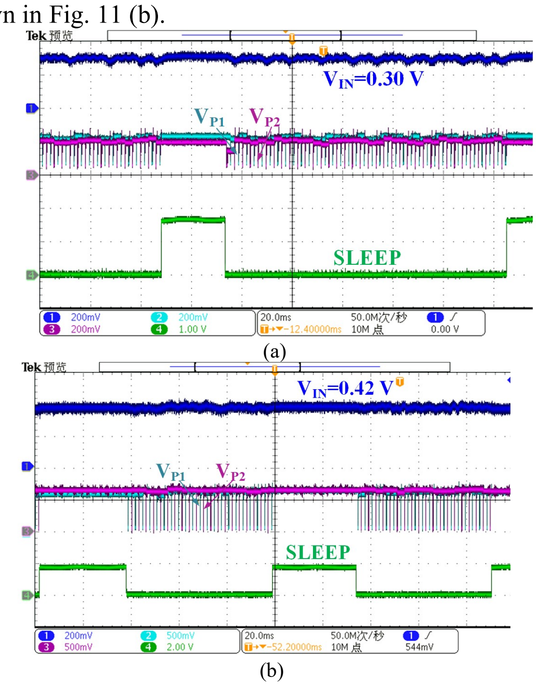


Fig. 11 Measured MPPT at (a) illumination of 5000 Lux and (b) 20 KLux

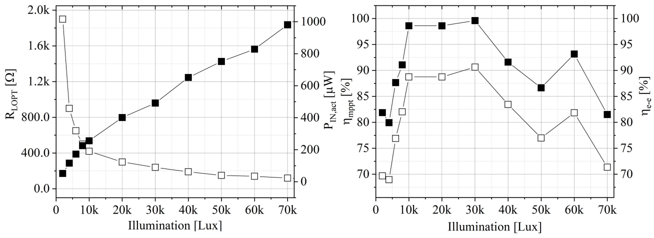
Fig. 12 (a) shows the measured maximum output power and optimal load resistance across 1,000–70,000 lux. Actual input power ($P_{IN,act}$) rises with illumination, while optimal load resistance (R_{LOPT}) decreases. Fig. 12 (b) plots MPPT efficiency (η_{mppt}) and end-to-end efficiency ($\eta_{e\text{-}e}$) versus illumination. The system reaches 99.6% peak η_{mppt} at 30k lux, while $\eta_{e\text{-}e}$ remains above 70% across the range, peaking at 90.6% under the same condition.

$$\eta_{mppt} = P_{IN,act} / P_{IN,\max} \tag{7}$$

where $P_{IN,\max}$ is the maximum available power of PV module.

$$\eta_{e\text{-}e} = P_{OUT} / P_{IN,act} \tag{8}$$

where P_{OUT} is the output power of the total energy harvester.



(a) (b)

Fig. 12 Measured (a) R_{LOPT} and P_{OUT} vs illumination and (b) η_{mppt} and η_{e-e} vs illumination

Table I compares the performance of the interface circuit proposed in this work with state-of-the-art designs. The proposed circuit implements a continuous perturb-and-observe MPPT scheme, achieving a peak power conversion efficiency of 90.6% and an MPPT efficiency of 99.6% over an input power range of 0.05–1.3 mW, while consuming only 2.04 µW of static power. Compared with existing solutions, the proposed interface circuit exhibits the lowest static power consumption and maintains outstanding peak conversion efficiency.

Table I. Performance comparison with other state-of-the-art works

	This work	TCASI'24 [9]	JSSC'24 [10]	TCASII'22 [11]	TPEL'21 [12]
Tech. [nm]	**180**	180	180	180	180
Topol.	**boost**	boost	boost	boost	boost
MPPT method	**P &O**	TVCB	Adaptive step	Hill climbing	P & O
Cont. MPPT	**Yes**	Yes	Yes	Yes	Yes
Inductor [µH]	**68**	10	NA	1	47
Freq. [KHz]	**10**	71	NA	250/125	NA
Area [mm^2]	**1.07**	0.365	1.17	1.4	0.95
Cold start	**Yes**	Yes	Yes	Yes	Yes
P_{IN} [mW]	**0.05-1.3**	28	33	119	0.1-3
P_{static} [µW]	**2.04**	2.64	5.47	1400	9.0
Max. η_{mppt} [%]	**99.6**	99.7	99.5	99.4	99.3
Max. $\eta_{E\text{-}E}$ [%]	**90.6**	NA	88	NA	94.7

IV. Conclusion

This work proposes a direct-sensing MPPT technique for photovoltaic energy harvesting. The input power from the PV cell is directly measured with a ripple-based current sensing and a time-domain multiplier, which significantly enhances the MPPT tracking speed and accuracy. A periodic sleep mode reduces power consumption while maintaining high MPPT efficiency. Operating from an ultra-low 180 mV input, the interface circuits achieve 99.6% peak tracking efficiency and 90.6% end-to-end efficiency. Compared to state-of-the-art solutions, it offers a superior balance between power consumption and tracking performance. Its integrated cold-start capability enables self-sustained operation, ideal for energy-autonomous IoT applications.

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